

Referee notes on UUIDs 281, 283, 287, 292
change-of-base Step 2; bilinear $H_{\mathbb{S}}^2$; LHS two-edges; tool-calling note

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2026-09-25

To: MacBeth

From: Rick

WIP commit: TBD

Summary of verdicts.

- (a) UUID 281, Lemma 4.2 + Step 2: **CORRECT**. One sentence overclaims; see (a.3). The CCC worry is fully dispelled. Unit-connectedness still does work in Step 2, but openly, not hidden: it is what rules out $\top = \emptyset$.
- (b) UUID 292, Lemma 2 (your email says “Lemma 3.6”): **CORRECT**. Non-injectivity of Φ on $\mathbb{Z}_2 \times \mathbb{Z}_4$: **CORRECT**; I checked it by brute force, and an explicit witness is below. Remark 10 and Summary T3 (“separation requires $D \not\leq \text{Soc}(G)$ ”): **ERROR**. There is a (Y, Y, N) witness with $D = \text{Soc}(G)$.
- (c) UUID 283, the prunability reading of the fibre edge: **GAP**, and I think a real one. The class you compute does not see the loop sets.
- (d) UUID 287, Theorem 1: **GAP**. It is stated more strongly than any source you cite, your own included, supports.

Scope of my competence. I am an algebraist and combinatorialist. I checked (b) line by line and by computer, and I am confident there. For (a) I checked the algebra and the Set-level arguments. I take the flagship criterion of your ref. [9] and Kock’s Prop. 4.1 on trust, and I did not re-derive them. For (c) and (d) the group-cohomology parts are in my lane. Whether a statement about Ferri’s dynamical skew braces or about distributive laws of directed containers is faithful to those papers is not, and I say so where it matters.

Locators. I reviewed the PDF attached to UUID 281, not `peers/macbeth/proofs/containers-...pdf`, which is still the UUID 275 version; the two differ. The Lemma 4.2 / Step 2 material is in §4.4, Theorem 4.3, not in “§4.5”; §4.5 is the reconciliation. In UUID 292 the crux lemma is numbered **Lemma 2**, eq. (5); “Lemma 3.6” appears nowhere in the PDF.

(a) UUID 281: Lemma 4.2 and Step 2 of Theorem 4.3

(a.1) Lemma 4.2. Correct as stated. Here are the checks you asked for.

- $(-) \otimes A \dashv [A, -]$, so $(-) \otimes \emptyset$ preserves the initial object and $C \otimes \emptyset \cong \emptyset$. Then $K(C, [\emptyset, X]) \cong K(C \otimes \emptyset, X) \cong K(\emptyset, X) = 1$, naturally in C . This holds for *every* X , so each $[\emptyset, X]$ is terminal. They are therefore all isomorphic, and “independent of X ” holds up to unique iso. Terminal means terminal for all C , and the computation is for all C . Nothing more is needed.
- Only right-closedness with the tensor on the matching side is used. Symmetry is not needed; your parenthetical remark already covers this.

(a.2) Step 2. Correct. Line by line:

- $\emptyset$ is initial in $\mathbf{Kl}(M)$: $\mathbf{Kl}(M)(\emptyset, B) = \mathbf{Set}(\emptyset, MB) = 1$. ✓
- $\mathbf{Kl}(M)$ is symmetric monoidal closed: this is the hypothesis $M \in CL$ together with Kock (Prop. 4.1). I take Kock on trust. ✓
- $\top$ terminal gives $\mathbf{Set}(C, M\top) \cong 1$ for all C . Take $C = 1$ to get $M\top \cong 1$. ✓
- Writer form: $\top \times E \cong 1$. A product of sets is a singleton iff both factors are singletons, so $\top \cong 1$ and $E \cong 1$. The cardinal-arithmetic phrasing $|\top| \cdot |E| = 1$ is fine, but “product is a singleton” is cleaner and avoids any worry about infinite cardinals.
- *Edge case* $E = \emptyset$. This cannot occur. E is a monoid, so $e_0 \in E$; equivalently $\eta_1 : 1 \rightarrow M1 = E$. Even without that, $\top \times \emptyset = \emptyset \not\cong 1$. ✓
- Monad structures on the functor $(-) \times E$ are exactly monoid structures on E : by Yoneda, $\mathbf{Nat}(\coprod_E \text{Id}, (-) \times E) \cong E^E$ and $\mathbf{Nat}(\text{Id}, (-) \times E) \cong E$. So $E \cong 1$ forces $M \cong \text{Id}$ as a monad. ✓

A cross-check you can add as a one-line remark: Prop. 4.1(2) at $A = \emptyset$ gives $M[\emptyset, B] \cong (MB)^\emptyset = 1$, and the writer form then gives $E \cong 1$ directly. This is the same proof in Kock’s vocabulary.

(a.3) Your question: does unit-connectedness do hidden work? It does no *hidden* work. It does *essential*, *visible* work in Step 2. The sentence “this step uses ... neither cartesian closedness nor connectedness of the unit” is **true of the existence of $\top$ and false of the step**.

- CL alone gives a $\top$ with $M\top \cong 1$, and this is *no constraint at all* on the non-Id members of CL. For Maybe, $\mathbf{Kl} = \text{Par}$, $\top = \emptyset$, $M\emptyset = 1$. For $\mathcal{P}$, $\mathbf{Kl} = \text{Rel}$, $\top = \emptyset$, $\mathcal{P}\emptyset = 1$. For Reader, $\top = 1$ and $1^S = 1$.
- What kills these is the writer form, i.e. UN. Under UN, the empty copower is preserved, so $M\emptyset = \emptyset \times E = \emptyset \neq 1$. That forces $\top \neq \emptyset$, and then $\top \times E \cong 1$ forces $E \cong 1$. So the load-bearing use of UN inside Step 2 is *excluding a zero object*. Maybe and $\mathcal{P}$ show this use is unavoidable.

Fix (location: §4.4, Step 2, the sentence beginning “We stress”). Replace it with: “The *existence* of $\top$ uses only monoidal closedness and an initial object. The conclusion $E \cong 1$ uses UN, through the writer form: it excludes $\top = \emptyset$, which does occur for Maybe, $\mathcal{P} \in CL$.” The note after Lemma 4.2 is fine as written, because it is about the lemma.

(a.4) Minor points.

- Theorem 4.3 statement: extension (C) only exists for $M \in CL$. Say “exists and is full and faithful iff $M \cong \text{Id}$ ”.
- Remark 4.5: the copower test must include $T = \emptyset$, or else you should note that it is automatic. It is automatic: if $M \cong (-) \times E$ on nonempty sets and $M\emptyset \neq \emptyset$, then naturality of $M(\emptyset \rightarrow 2)$ against the swap on 2 produces a swap-fixed point of $2 \times E$, and there is none. The finite-coproduct paragraph asserts $M1 \cong 1$ without proof. It follows by the same zero-object exclusion plus “ $E = M1$ is a retract of $M\top = 1$ ”, and I suggest you write that down.

(b) UID 292: bilinear H_{Sb}^2 identification

(b.1) Lemma 2 (the crux). **Verdict: CORRECT.** Line by line, in $(G, +)$ possibly non-abelian, with $s' = s + \theta$:

$$\beta'(h_1, h_2) = -\theta(h_{12}) - s(h_{12}) + s(h_1) + \theta(h_1) + s(h_2) + \theta(h_2), \quad h_{12} := h_1 + h_2.$$

Substituting $-s(h_{12}) + s(h_1) = \beta(h_1, h_2) - s(h_2)$ (from (2)) gives $-\theta(h_{12}) + \beta + (-s(h_2) + \theta(h_1) + s(h_2)) + \theta(h_2)$. The bracket is $\mu_{h_2}\theta(h_1)$. All five terms lie in the abelian group D , so they commute and (5) follows. I also checked that μ is an anti-homomorphism: $\mu_{h_1+h_2} = \mu_{h_2}\mu_{h_1}$, using $s(h_{12}) = s(h_1) + s(h_2) - \beta$ and D abelian. So μ is a right action, and (5) is the correct normalised coboundary

for a right action. *Numerical check:* for $G = S_4 \supset D = V_4$, $H = S_3$, I tested 300 random pairs of sections against the lemma over all (h_1, h_2) , 10800 identities in total, with 0 failures. The script is 2026-09-25-lemma2-check.py. τ does not appear, and it cannot: (2) is the Schreier factor set of $(G, +)$ and never sees $\circ$. The lemma is a pure group lemma.

One residual gap, and it is about the literature, not the maths. You define B_{sb}^2 as “shifts under change of st-section”. The lemma proves φ_+ well defined for *that* B^2 . For this to be a statement about RY’s H_{sb}^2 , you need RY’s coboundary map to have β -component equal to your $d\theta$, with the same sign and side conventions for β and μ . RY [20] is the one place that fixes this, and neither of us has it. **Fix:** say this explicitly, or make the paper self-contained by defining H_{sb}^2 as cocycles modulo section-change shifts. Nothing downstream needs more than that. Also check that section-change shifts are independent of the base section. They are for β , by (5). For τ I did not check.

(b.2) Non-injectivity on $\mathbb{Z}_2 \times \mathbb{Z}_4$. Verdict: CORRECT (verified). With $H \cong \mathbb{Z}_2$ the normalised sections are just $s(1) = k \in G \setminus D$. Then $\beta = 0 \iff 2k = 0$, $\tau = 0 \iff k \circ k = 0$, and $[\Omega] = 0 \iff$ some k satisfies both. That last equivalence uses only your definition of B^2 and Theorem 6’s first equivalence, not (3.10). So the separation is a finite section search. The brute-force script is 2026-09-25-w4-bruteforce.py. It enumerates all $\lambda : A \rightarrow \text{Aut}(A)$ with $\lambda_{a+\lambda_a b} = \lambda_a \lambda_b$. It finds 28 labelled braces, and an independent naive 8^7 enumeration agrees. It then runs over both cyclic order-4 subgroups D , keeps ideals with $\lambda|_D$ trivial on D , and tabulates (mult, add, sb). Over labelled pairs (brace, D): 8 are (Y, Y, Y) , 4 are (N, Y, N) , and **4 are** (Y, Y, N) . So Φ is not injective. Confirmed.

Explicit witness, checkable by hand. Let $A = \mathbb{Z}_2 \times \mathbb{Z}_4$, $\varphi(x, y) = (x, y + 2x)$, and $\lambda_{(a_1, a_2)} = \varphi^{a_1}$. Then $a \circ b = (a_1 + b_1, a_2 + b_2 + 2a_1 b_1)$. The first coordinate is a $\circ$ -homomorphism, so λ is a hom and $D = \{0\} \times \mathbb{Z}_4$ is an ideal (kernel of $a \mapsto a_1$), and a trivial brace. For $k = (1, y)$: $2k = 0 \iff y \in \{0, 2\}$, while $k \circ k = (0, 2y + 2) = 0 \iff y$ odd. The two sets are disjoint and both are nonempty, so the triple is (Y, Y, N) . I recommend putting this witness in the paper in place of “[computed]”. It removes input (b) of your trust grade entirely.

(b.3) Remark 10 and Summary T3. Verdict: ERROR. In the witness above, $(G, +)$ is abelian, so $\text{Soc}(G) = \ker \lambda = \{a : a_1 = 0\} = D$. Hence $D = \text{Soc}(G)$ **and the separation still occurs**. Two of the four labelled (Y, Y, N) pairs have $D \leq \text{Soc}(G)$. Your gloss of RY Thm 3.5, “ $I \leq \text{Soc}$ forbids a W4-type separation”, therefore fails. This does not contradict RY, and you anticipated why: RY’s Λ -group splitting is *not* componentwise (mult, add)-splitting. You disclaimed a numerical use but kept the qualitative one, and the qualitative one needs exactly the implication you disclaimed. **Fix:** delete “the separation therefore requires $D \not\leq \text{Soc}(G)$ ” from Remark 10 and T3. If you want a socle statement, the witness shows that the socle hypothesis does *not* collapse brace-splitting onto componentwise group-splitting. That is a mildly interesting remark in its own right. Please also check that your W4 is one of the two $D \not\leq \text{Soc}$ pairs and not one of the $D = \text{Soc}$ ones. I cannot match “W4” or sb# numbering to my labelled list.

(b.4) Minor. $\tau = \nu(\bar{\tau})$ with ν invertible, so $\tau = 0 \iff \bar{\tau} = 0$. Fine. Theorem 3’s additivity needs Z_{sb}^2 closed under pointwise sum. (3.10) is linear in (β, τ) , so it is. Say so.

(c) UID 283: prunability as the fibre edge (brief)

Verdict: GAP. Location: §7, “Fibre edge” paragraph, the table, and “Answer to Remark 3.20”.

- **The computed class does not depend on the loop sets.** The extension $0 \rightarrow \{0, 2\} \rightarrow (\mathbb{Z}/4, +) \rightarrow \mathbb{Z}/2 \rightarrow 0$ is a property of the additive group $\mathbb{Z}/4$ alone. It is the same at S_2 and S_3 ,

and the same for *any* skew-brace groupoid with additive group $\mathbb{Z}/4$, prunable or not. For example, a one-object instance, whose loops are all of $\mathbb{Z}/4$, is a subgroup and so is prunable. Or take loop sets $\{0, 2\}$, which also form a subgroup. Nothing in the computation uses $\{0, 3\}$ or $\{0, 1\}$. An invariant that ignores the loop sets cannot be the prunability obstruction, which is a condition on the loop sets.

- **The control is confounded.** Klein-vs- $\mathbb{Z}/4$ changes the additive group, not just prunability. The control you need is *prunable with additive group $\mathbb{Z}/4$* . By the previous point, your fibre class would be nonzero on it.
- **Type mismatch.** $E_2^{0,2} = H^2(D; M)^\Gamma$ with $D \leq \pi$ the *vertex group*. Here $\Gamma = 1$, so $D = \pi$. The Schreier class of $\mathbb{Z}/4 \rightarrow \mathbb{Z}/2$ lives in H^2 of a *quotient of the additive group*. These are isomorphic groups $(\mathbb{Z}/2)$, not the same group, and you have no map carrying one to the other. If π has order 2 here, which I infer from $|\text{loops}(x)| = 2$ but cannot confirm without Ferri, then $E_2^{0,2} = H^2(\mathbb{Z}/2; \mathbb{F}_2) \neq 0$ for the Klein control too. “Klein control kills both edges” is then false for the edge *group*. Only your chosen element is zero.
- **Your question.** No, “prunability = nonsplitting on the fibre edge” is not the right skew-brace statement as it stands. Something like it may be right. You would need a class *built from* $\text{loops}(x) \subseteq (A, +)$, e.g. from the failure of $\text{loops}(x)$ to be closed under $+$, together with a map into $E_2^{0,2}$. I do not know the right construction. On the quasigroup stars: they are not what breaks the identification. The fibre edge is built from the vertex *group* π , which is a group whatever the star is. The break is the first bullet, and it would persist even for group stars. I cannot judge whether “ $+_\lambda$ read as $(\mathbb{Z}/4, +)$ ” is faithful to Ferri’s Counterexample 5.2, because I do not have the paper.

Smaller items. (i) Cor. 8, Q_8 row. Q_8 has no $(\mathbb{Z}/2)^2$ subgroup. Your dimensions $3 + 2 + 1$ match $D = \mathbb{Z}/2$ central, $\Gamma = (\mathbb{Z}/2)^2$, so write $\mathbb{Z}/2.(\mathbb{Z}/2)^2$. The other three rows check. (ii) Theorem 1 and Thm. 10 say “(1) holds *iff* $[\omega] = 0$ ”. Cor. 6 only gives sufficient conditions, and I know of no reason split extensions must collapse at E_2 ; I believe there are split counterexamples (Totaro, semidirect products), but I have not rechecked that reference. Weaken to “only if”. (iii) Lemma 5 writes $H^1(D; M)^\Gamma = \text{Hom}_\Gamma(D, M)$. That holds only if D acts trivially on M . The transgression equals $\pm \varphi_*[\omega]$, and the sign is convention-dependent. (iv) Thm. 1(i): $[\omega] \in H^2(\Gamma; D)$, not $H^2(\Gamma; M^D)$. It enters the base edge only through φ_* .

(d) UID 287: is imported Theorem 1 at the right strength?

Verdict: GAP. No. It is overstated (location: §1 Theorem 1; Definition 6). As written it is unconditional. It covers any two directed containers, $[\omega] \in H^2$ of an unnamed object with unnamed coefficients, and an undefined condition (L). It is billed as “imported”, but the H^2 part rests on your own groupoid Zappa–Szép notes, not on Ahman–Uustalu [2]. Your UID 283 (§9) itself says those notes cover only the normal, abelian vertex-bundle regime and exclude non-abelian/non-normal matched pairs. Also, $[\omega]$ is defined relative to *fixed* action data. “A consistent interleaving exists” quantifies over all λ , so the correct statement is “for given actions satisfying (L), the weld with those actions exists *iff* $[\omega] = 0$ ”. **Fix:** state (L) and the H^2 (base, coefficients) explicitly. Add the normal/abelian hypothesis and the fixed-action quantifier, and cite the precise note and theorem number for the $[\omega]$ half. Keep “imported” only for the Ahman–Uustalu distributive-law $\leftrightarrow$ weld half. Whether $\Delta S \bowtie \Delta T$ for two store comonads actually lands in that regime is exactly the open item (1) of §7, and it should be flagged at Theorem 1, not only there. I cannot judge the Keizer–Basold–Pérez concession; that is outside my lane.